

PSY310- Lab in Psychology

Section 1

Lab Report 1- Stroop Effect

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Student Details

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Git hub link:<https://github.com/aviks-afk/PSY-310-LAB-IN-PSYCHOLOGY.git>

Introduction

Here Stroop Effect will be seen in how automatic and deliberate cognitive processes take place in the interference with one another. It is observed in cases when the subjects are requested to name the ink color of a word which names a colour. Incongruence between the word meaning and the ink color (the word RED printed in blue ink) yields slower and less accurate responses as compared to conditions of congruence (word RED printed on red ink).

Such effect can be explained by the automatization of reading that disrupts the effortful process of colour naming. The Stroop therefore offers a clue into the ability to exert selective attention and inhibition as well as cognitive control.

This research was done to compare the correctness and the speed of response between the congruent and the incongruent condition. It was the hypothesis that:

The second is because accuracy will be greater in the congruent than in the incongruent condition.

Response will be quicker in congruent than in incongruent condition.

Method

Participants

There were four participants (two males and two females), of ages 19 to 21 years, included in the selection from Ahmedabad University through convenience sampling. It was ensured that the participants had an informed consent, confidentiality was guaranteed, and the right of withdrawing was ensured.

Apparatus

The experiment was performed in PsychoPy v2024.2.1, on a laptop (13.3 display, 2560 1600 pixels, 60 Hz refresh). Responses were captured by use of keyboard keys.

Stimuli and Stimuli

Two conditions were involved in a within-subjects design.

Congruent condition: The word meaning and ink color match.

Incongruent condition: The meaning of the words and colour of ink did not correlate.

Stimulus was word presented centrally in colored ink. The subjects were asked to answer the ink color and thus not think about the word meaning.

Procedure

In each condition, the participants were then given trials after practice phase. The stimuli always started with a fixation cross lasting 500 ms. The stimulus was maintained until the time of the response or maximum of 2 seconds. To assess real-life performance, RTs and accurate responses were automatically recorded on PsychoPy.

Results

Accuracy

Participant	Congruent (%)	Incongruent (%)	Average (%)
P1	100	70	85
P2	100	70	85
P3	100	65	82.5
P4	100	68	84
Average	100	68.25	

Interpretation: Accuracy was perfect (100%) in the congruent condition, but dropped to an average of 68.25% in the incongruent condition.

Average Response Time

Condition	P1	P2	P3	P4	Average
Congruent	1.0748	1.0602	0.9214	0.7748	0.9578
Incongruent	1.1774	1.2516	1.0744	0.6994	1.0507

Interpretation: Participants responded faster in the congruent condition ($M = 0.9578$ s) compared to the incongruent condition ($M = 1.0507$ s).

Discussion

The findings back the hypotheses. The results indicated a significantly more accurate response time and response time in congruent trials than in incongruent trials, due to the classic Stroop effect. This is an indication of interference between the automatic reading process and the controlled color naming process.

The results parallel the findings of Stroop (1935) and other subsequent reviews that indicated incongruent trials use more cognitive control resulting in less accurate and slower responses. Limitations include small sample size, which does not allow generalizing the result, and absence of neutral trials (e.g. non-color words in colored ink) that could serve as an analogue against which to compare outcomes.

In follow up studies, samples may be larger and more representative or gender, age, and the emotional nature of the task may be explored.

References

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